1 ECE 188 Notes

1.1 Non-Linearity and Distortion

1.1.1 Linear Time Invariant

A system is linear, time-invariant if

$$\begin{aligned} x(t) \to y(t) \quad &\Rightarrow \quad ax(t) \to ay(t) \\ x(t) \to y(t) \quad &\Rightarrow \quad x(t-\tau) \to y(t-\tau) \end{aligned}$$

Linear systems can generate frequency components that do not exist in the input.

1.1.2 Simple Nonlinear Model

Static (Memory-less) model (nonlinear models are mostly compressive and cause Harmonic distortion, etc)

$$y(t) = \alpha_0 + \alpha_1 x(t) + \alpha_2 x^2(t) + \cdots$$

1.1.3 Desensitization and Blocking

For signal with a larger interferer, ω_1 and ω_2, nonlinear system causes and interference called Intermodulation.